Tithi Kotasthane

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RESEARCH INTERESTS

Biology is not static. I am fascinated by how cells sense and integrate changing cues to regulate growth, development and adaptation. My current research in *Drosophila melanogaster* investigates whether hypoxia signalling can be regulated beyond the classical oxygen-sensing pathway in response to nutritional cues. I am particularly interested in how the timing and context of a signal shape the response.

EDUCATION

Integrated PhD (iPhD), IISER Thiruvananthapuram Jul 2024 - present
3rd year. CGPA: 9.56/10 (up to Year 2)

B.Sc. Cell and Molecular Biology, The Maharaja Sayajirao University of Baroda 2021 - 2024
CGPA: 9.08/10

Class XII, Gujarat Secondary and Higher Secondary Education Board 2021
Percentile rank: 98.6
Awarded the INSPIRE Scholarship for academic performance (top 1% in the Class XII board examination)

RESEARCH EXPERIENCE

Master's Thesis, DREAM Lab, IISER Thiruvananthapuram Jan 2026 - present
Supervisor: Dr. Jishy Varghese
Project: *Deciphering the role of hypoxia signalling in growth, development and nutritional homeostasis in Drosophila melanogaster*

- Investigating how the hypoxia pathway integrates nutritional and metabolic stress signals and contributes to growth, survival and developmental physiology in *Drosophila melanogaster*.
- Examining hypoxia pathway regulation under starvation and hypoxia, and its interaction with insulin and ecdysone signalling across development and adulthood.
- Using genetic manipulation of the hypoxia pathway in adult flies to test whether its activity influences physiological responses and survival under metabolic stress.

Research Rotation, IISER Thiruvananthapuram Aug - Sep 2025
Supervisor: Dr. Nisha N. Kannan

- Trained in *Drosophila* chronobiology, with exposure to interactions between circadian regulation and environmental inputs, including nutrition.

Research Rotation, IISER Thiruvananthapuram Jun - Aug 2025
Supervisor: Dr. Sabari Sankar Thirupathy

- Investigated transcriptional regulation in *Bacillus subtilis*, focusing on transcription-replication conflicts in head-on and co-directional orientations.
- Gained experience in bacterial culture, minimal media preparation and flow cytometry-based analysis.

Summer Research Internship, IISER Pune May - Jul 2024
Supervisor: Dr. Girish Ratnaparkhi

- Studied the role of serine hydroxylases in *Drosophila* embryonic development.

Summer Research Internships, The Maharaja Sayajirao University of Baroda 2022 - 2023
Supervisors: Dr. Devashri Gajjar, Dr. Ankita Salunke

EXAMINATIONS

- **DBT-BET** (Biotechnology Eligibility Test): Qualified for JRF
- **IIT-JAM** (Joint Admission Test for M.Sc.): AIR 159
- **GAT-B** (Graduate Aptitude Test - Biotechnology): Qualified
- **JGEEBILS** (Joint Graduate Entrance Examination for Biology and Interdisciplinary Life Sciences): Qualified

TECHNIQUES

***Drosophila* and physiological assays**	*Drosophila melanogaster* handling and maintenance, larval synchronisation and developmental staging, tissue dissection (ovary, brain), feeding assays, starvation and survival assays, activity monitoring, genetic perturbation (RNAi, GeneSwitch, overexpression)
Molecular biology and biochemistry	RNA extraction and cDNA synthesis, DNA extraction and plasmid isolation, PCR, gel electrophoresis, qPCR, basic chromatography, bacterial culture, antimicrobial susceptibility assays
Microscopy and imaging	Confocal microscopy, fluorescence imaging, sample preparation, basic image analysis
Cellular and data analysis	Flow cytometry sample preparation and basic analysis, Python (basic data analysis and visualisation)

OTHER PURSUITS

- **Off-the-Bench Initiative**, IISER Thiruvananthapuram. A student-led initiative promoting scientific discussion beyond routine laboratory work
- Participated in molecular biology and proteomics workshops and conferences
- **Mimamsa** (IISER Pune science quiz): Zonal-level topper
- **Kathak**: Six years of formal training in Indian classical dance